Grade Book Project

Electronic Gradebook System Technology Project Overview

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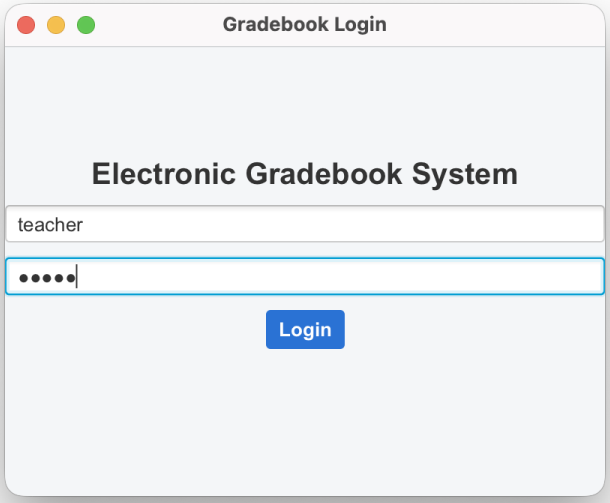
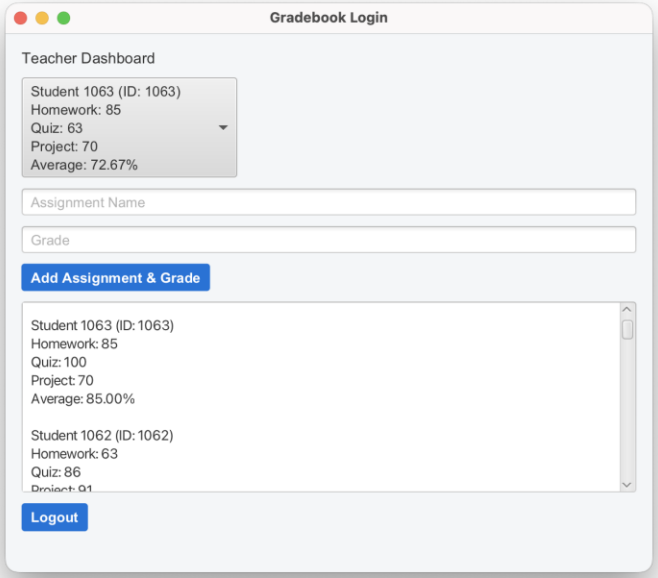
Project Overview

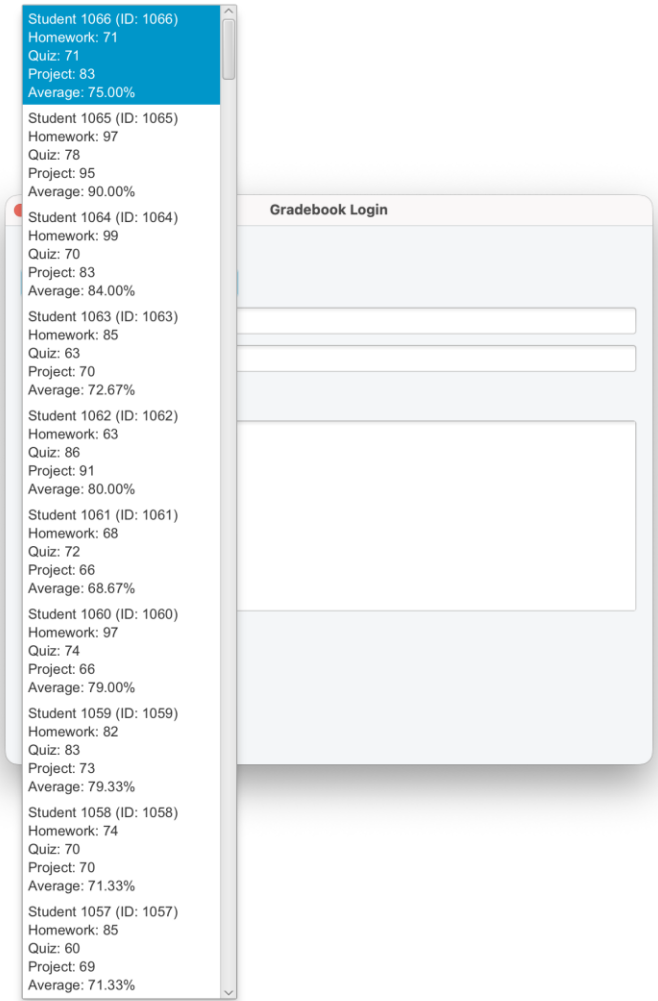
- • JavaFX-based electronic gradebook application
- • Supports teacher and student logins
- • Allows grade entry, calculation, and display
- • Designed for usability and academic record management

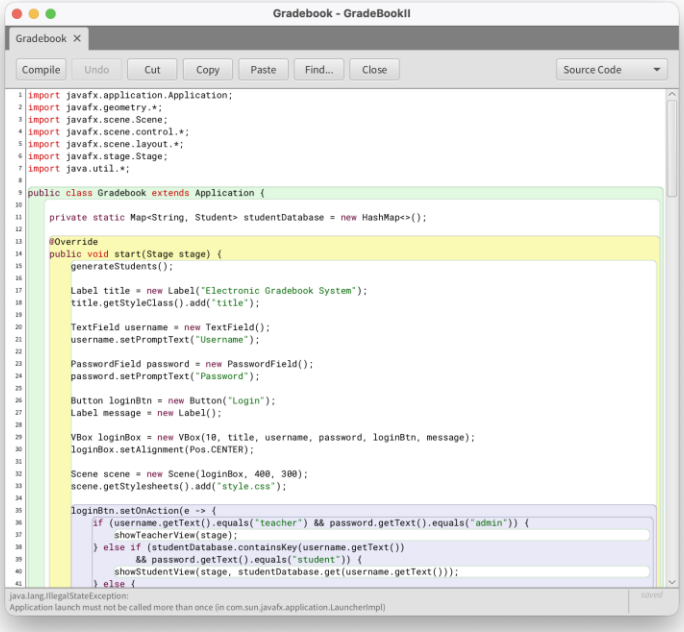
Login Interface





Teacher Dashboard & Grade Management





Code Implementation

Gradebook - GradeBookII

Gradebook

Compile | Undo | Cut | Copy | Paste | Find... | Close | Source Code

```java
        String assignment = assignmentField.getText();
        String gradeText = gradeField.getText();

        if (selected != null && !assignment.isEmpty() && !gradeText.isEmpty()) {
            try {
                double grade = Double.parseDouble(gradeText);
                selected.addAssignment(assignment, grade);

                // Refresh display
                area.clear();
                for (Student s : studentDatabase.values()) {
                    area.appendText(s.toString() + "\n");
                }

                assignmentField.clear();
                gradeField.clear();

            } catch (NumberFormatException ex) {
                area.appendText("Invalid grade format.\n");
            }
        }
    });

    Button logout = new Button("Logout");
    logout.setOnAction(e -> start(stage));

    VBox root = new VBox(
            10,
            new Label("Teacher Dashboard"),
            studentBox,
            assignmentField,
            gradeField,
            addGradeBtn,
            area,
            logout
    );

    root.setPadding(new Insets(15));

    Scene scene = new Scene(root, 600, 500);
```

java.lang.IllegalStateException:
Application launch must not be called more than once (in com.sun.javafx.application.LauncherImpl)

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Gradebook - GradeBookII

Gradebook

Compile | Undo | Cut | Copy | Paste | Find... | Close | Source Code

```java
    int homework, quiz, project;

    public Student(int id, String name, int hw, int qz, int pr) {
        this.id = id;
        this.name = name;
        this.homework = hw;
        this.quiz = qz;
        this.project = pr;
    }

    // ↖ Added method to fix compile error
    public void addAssignment(String assignment, double grade) {
        switch (assignment.toLowerCase()) {
            case "homework":
                homework = (int) grade;
                break;
            case "quiz":
                quiz = (int) grade;
                break;
            case "project":
                project = (int) grade;
                break;
            default:
                // Ignore invalid assignment names
                break;
        }
    }

    public double average() {
        return (homework + quiz + project) / 3.0;
    }

    @Override
    public String toString() {
        return name + " (ID: " + id + ")\n" +
                "Homework: " + homework + "\n" +
                "Quiz: " + quiz + "\n" +
                "Project: " + project + "\n" +
                "Average: " + String.format("%.2f", average()) + "%\n";
    }
}
```

java.lang.IllegalStateException:
Application launch must not be called more than once (in com.sun.javafx.application.LauncherImpl)

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Project Timeline

- Planning & Design: January 17, 2026 - January 21, 2026
- Initial JavaFX UI Development: January 23, 2026
- Core Logic & Data Structures: January 23, 2026
- Testing & Refinement: January 24, 2026 - January 30, 2026
- Final Submission & Presentation: February 3, 2026